

Market Regime Clustering for Sector Rotation Strategy using GMM

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Problem Statement

This paper proposes a dynamic sector rotation strategy using unsupervised learning to identify market regimes from macroeconomic and financial data. Gaussian Mixture Models (GMMs) are trained on engineered features such as daily sector returns, volatility measures, and yield spreads. The model attempts to detect structural shifts in the financial time series, which is then used for portfolio reallocation among sectors to enhance returns.

Contents

1	Introduction	2
2	Data Description	2
2.1	Data Source	2
2.2	Feature Engineering	3
3	Methodology: Gaussian Mixture Model	5
4	Regime Interpretation	7
4.1	Macroeconomic Analysis per Regime	7
4.2	Sector ETF Returns per Regime	9
5	Sector Rotation Strategy	10
6	Evaluation and Final Results	11
7	Conclusion	12

1 Introduction

Market behavior is not stationary, and they can exhibit different behavioral patterns over time, with periods of stability, growth, and volatility. These periods are known as market regimes. Being able to identify these regimes in real time allows investors to adapt their strategies accordingly. A sector rotation strategy reallocates capital into outperforming sectors during favorable regimes while avoiding those expected to underperform. But by the time it becomes obvious which market regime we are in, the best opportunity to reallocate funds may have already passed.

Gaussian Mixture Models (GMMs) can help model the joint distribution of financial and macroeconomic features as a mixture of multivariate Gaussian distributions. GMM is considered a soft clustering technique which assigns likelihoods to different market conditions rather than hard labels. This is beneficial for financial time series modeling, where transitions between regimes are generally gradual rather than abrupt. Additionally, deviations from the learned mixture distributions over time can signal potential regime shifts, as the observed data begins to diverge from the statistical patterns captured by the original GMM components. By monitoring these deviations, it becomes possible to detect an incoming regime change earlier and proactively adjust sector allocations, allowing the rotation strategy to respond dynamically before market conditions fully transition.

2 Data Description

2.1 Data Source

This study will be relying on financial and macroeconomic data from two primary sources: Yahoo Finance and the Federal Reserve Economic Data (FRED) API. The features from both sources are specifically chosen to reflect investor sentiment, economic outlook, and actual risk conditions between the years 2000 to 2024

From Yahoo Finance:

- **Sector ETFs:** Daily adjusted closing prices for XLK (Technology), XLF (Financials), XLE (Energy), XLV (Healthcare), XLY (Consumer Discretionary), XLU (Utilities), XLI (Industrials), XLB (Materials), IYR (Real Estate), and XLP (Consumer Staples)
- **Market Index:** The S&P 500 Index (SPY)
- **Volatility Index:** The VIX, also known as the “fear index,” measures how much investors think the stock market will move in the near future

From the FRED API:

- Quarterly Gross Domestic Product (GDP) levels
- Monthly Consumer Price Index (CPI) as a measure of inflation

- **Unemployment Rate:** Monthly unemployment statistics
- **Fed Funds Rate:** The Federal Reserve’s benchmark interest rate
- **Treasury Yields:**
 - 10-Year Treasury Constant Maturity Rate
 - 2-Year Treasury Constant Maturity Rate
 - 3-Month Treasury Rate
- **Consumer Sentiment:** University of Michigan Consumer Sentiment Index

Note, since several FRED macroeconomic indicators are reported at monthly or quarterly frequencies, these features were forward-filled to align with the rest of the dataset.

2.2 Feature Engineering

Gaussian Mixture Models (GMM) assumes that features are stationary. A stationary time series means that the series fluctuates around a fixed mean with a stable variance, like a Gaussian distribution. However, raw metrics like GDP and CPI values are reported as cumulative levels over time, as per Figure 1 below:

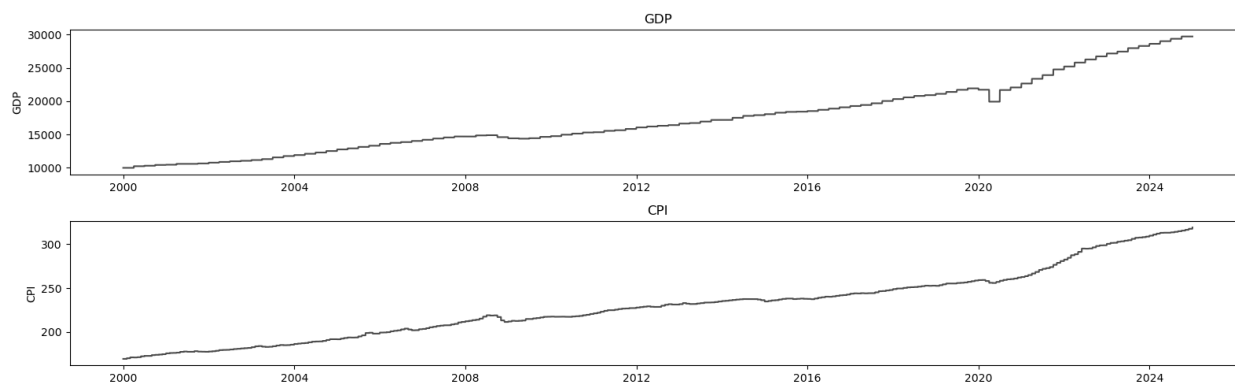


Figure 1: Raw GDP and CPI (2000 to 2024)

As observed above, both metrics naturally trend upward over time. Thus their magnitude could easily influence other features in the model that are stationary, such as RSI and yield spreads. To get around this, these metrics are transformed into rate-based features that reflect daily percentage change, thus making them stationary (Figure 2).



Figure 2: Percentage Changes for GDP and CPI

This same logic applies to each of the sector ETFs’ daily adjusted close prices, as well as the S&P 500 index. As such, the levels are also converted into daily percentage returns so that they can be equally compared in terms of volatility and direction. This prevents the GMM clustering from being skewed by the different price levels of each ETF.

To anticipate and weather out unfavorable regimes, a “risk-free” cash return feature is also engineered to create a proxy for holding short-term cash. This is simulated by using the 3-month Treasury yields obtained from FRED, and it represents how much would be earned on a daily basis if capital was converted into cash equivalents.

To capture the trend of the market, the **Simple Moving Average (SMA) Ratio** is calculated. This is the ratio between the 50-day and 200-day SMAs of the S&P 500 index on day t :

$$\text{SMA Ratio}_t = \frac{\text{SMA}_{50,t}}{\text{SMA}_{200,t}} \quad (1)$$

Additionally, an **SMA Difference** is also calculated, which is simply the difference between the 50-day and 200-day SMAs of the S&P 500 index. This difference provides the model with a direct measure of the momentum gap between short-term and long-term trends.

The **Relative Strength Index (RSI)** is another engineered feature, which helps quantify the strength of recent price movements by comparing the size of gains to losses over a fixed period. In this study, RSI is computed using a 14-day simple moving average of gains and losses derived from daily changes in the S&P 500 index on day t :

$$\text{RSI}_t = 100 - \frac{100}{1 + \frac{\text{Average Gain}_t}{\text{Average Loss}_t}} \quad (2)$$

Finally, **yield curves** are engineered from the difference in interest rates between long-term and short-term Treasury bonds. The first curve is based on the spread between 10-year and 2-year bonds. The second curve is compares 10-year and 3-month bonds. For the GMM

model, the former captures market expectations about long-term versus short-term growth, while the latter captures central bank policy shifts.

3 Methodology: Gaussian Mixture Model

With all the relevant features engineered, the data is now ready for clustering. Again, the objective is to not only identify, but to detect earlier, the distinct market regimes based on the combined behavior of these features. To achieve this, the Gaussian Mixture Model (GMM) is employed.

GMM is a probabilistic clustering algorithm that segments data based on the assumption that it is generated from a mixture of multivariate Gaussian distributions. Unlike hard clustering methods, GMM assigns soft probabilities to each data point. This uncertainty or overlap between clusters was chosen specifically for this study due to its ability to model soft transitions. In financial time series, market regimes tend to evolve gradually rather than shift abruptly. If there are n regimes, the model assigns a probability distribution over all n clusters for each observation, indicating the likelihood of that day belonging to each regime. These probabilities can then be used to make decisions in portfolio allocation. The GMM is mathematically represented as:

$$p(\mathbf{x}) = \sum_{k=1}^K \pi_k \mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k) \quad (3)$$

Where:

- $\mathbf{x} \in \mathbb{R}^d$ is a feature vector on a given day, composed of macroeconomic and financial return-based indicators
- K is the number market regimes
- π_k are the mixing coefficients, representing the prior probability of being in regime k , with $\sum_{k=1}^K \pi_k = 1$
- $\boldsymbol{\mu}_k$ is the mean vector for regime k .
- $\boldsymbol{\Sigma}_k$ is the covariance matrix for regime k that captures the variance and correlation of features within that regime.
- $\mathcal{N}(\cdot)$ indicates the multivariate normal distribution.

Before fitting the GMM, the features are standardized to ensure they are on the same scale using z-score normalization. This step is crucial because GMMs are sensitive to feature magnitudes, such that features with larger ranges can easily dominate the clustering.

And to further reduce noise, Principal Component Analysis (PCA) is applied to the scaled features. Essentially, PCA projects the feature set into a lower-dimensional space while capturing the highest variance in the data. This ensures that the most informative information

is kept while removing multicollinearity or features with low variance. In this study, the number of principal components chosen preserves 95% of the total variance. The resulting transformed data is then passed into the GMM for clustering.

The number of components is set to $K = 5$, corresponding to five market regimes. Through empirical trials, the number of regimes was determined to be 5 because it offered a good balance between the risks of overfitting and under-segmentation, resulting in clusters that are economically meaningful and interpretable.

The GMM is trained using the Expectation-Maximization (EM) algorithm, which estimates the parameters π_k, μ_k, Σ_k that maximizes the likelihood of the data under the mixture of multivariate Gaussian distributions. For each day, the model computes posterior probabilities belonging to each regime and assigns that day to the most likely regime.

However, financial time series are inherently noisy, and regime probabilities can fluctuate day to day due to rapid variations in the features. To prevent the model from overreacting to short-term noise, a smoothing rule is applied such that a regime change is only confirmed if the same new regime is detected in five consecutive days in a row. In other words, the process checks whether a moving window of length $m = 5$ contains identical regime labels. If so, the transition is accepted, otherwise, the previous regime is maintained.

The result of the GMM clustering process is a smoothed sequence of regime labels over time, where each day is assigned to one of the five identified regimes. This sequence captures structural shifts in market behavior, constrained by the 5-day confirmation rule to avoid reacting to short-term noise. Figure 3 illustrates the full timeline of regime assignments using color-coded bands:

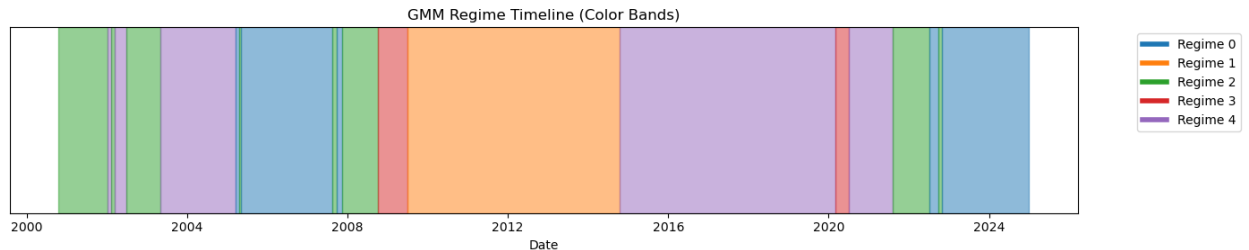


Figure 3: GMM Regime Timeline (2000 to 2024)

It is also possible to overlay the classified regimes onto the S&P 500 index, as per Figure 4 below. The index is displayed as a black line, while each regime is highlighted using colored scatter points that correspond to the same color legend used in Figure 3.

This visualization speaks to how the GMM-detected regimes relate to actual market dynamics experienced during this timeline. Sharp downturns such as the 2008 financial crisis and the early 2020 COVID-19 crash align closely with high-volatility regimes. In contrast, bull markets and recoveries are marked by regimes associated with strong momentum:

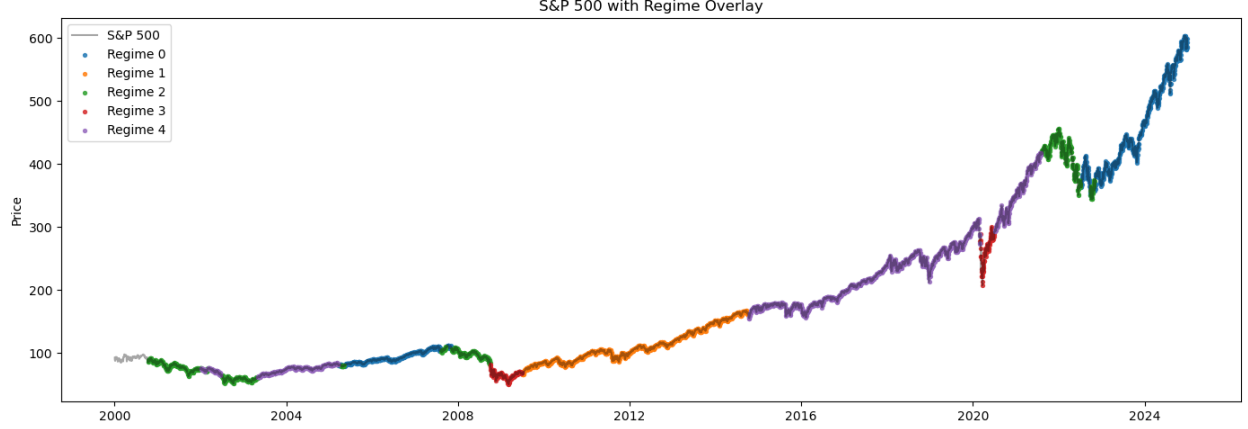


Figure 4: S&P 500 Index with Regime Overlay (2000 to 2024)

Note: It is important to note that the S&P 500 index is used here solely as an illustrative example. The GMM classification was performed using a general set of macroeconomic, volatility, and sector-specific features.

4 Regime Interpretation

The regime overlay on the S&P 500 index provides a visual (but basic) confirmation that the GMM model successfully distinguishes different market conditions. However, to fully understand the economic characteristics of each detected regime, a more detailed analysis is necessary. And by doing so, each regime can then be properly labeled. To do that, each regime is independently analyzed based on the *median* levels of the various macroeconomic indicators and their respective sector returns.

4.1 Macroeconomic Analysis per Regime

Regime	RSI	VIX	SMA Ratio	SMA diff	YC 10Y-2Y	YC 10Y-3M
0	58.857016	14.110000	1.039344	4.892981	-0.08	-0.320
1	58.258416	17.580000	1.056624	6.676443	2.25	2.580
2	47.849902	25.059999	0.962449	-3.616726	1.22	1.760
3	52.285687	42.250000	0.887144	-13.953241	1.85	2.575
4	57.449112	15.650000	1.041193	6.041144	1.05	1.540

Table 1: Median Macroeconomic Indicators Across Regimes

Using the median values in Table 1, the following labels and summaries for each GMM-clustered regime can be extrapolated:

Regime 0: Moderate Expansion

Regime 0 indicates stable upward momentum based on an observed RSI of 58.85 and low VIX levels of 14, indicating calm market conditions. The SMA_Ratio is slightly above 1, and the SMA_diff is positive, which suggests positive momentum without signs of overheating. However, the yield curve is slightly inverted (-0.08) and the 10Y-3M spread is also negative (-0.32). This implies the economy is late cycle, where the market is still strong but bond markets are hinting at a future slowdown or policy mistake. Essentially, this regime represents a steady-growth environment but underlying caution.

Regime 1: Bullish Risk-On

Regime 1 represents a strong bullish environment where the RSI remains high at 58 and the SMA_diff increases to 6.7, indicating stronger short-term momentum. Both the 10Y-2Y and 10Y-3M spreads are steeply positive (2.25 and 2.58 respectively), indicating optimism about future growth. Referring back to Figure 3, this cluster aligns with known expansion phases where monetary policy was loose and markets were risk-on.

Regime 2: Choppy Recovery

Regime 2 shows mixed signals. RSI levels are moderate at 48 with a slightly negative SMA_diff (-3.6), which indicates technical weakness and no clear trend. The VIX is higher at 25, pointing to more uncertainty and choppy market conditions. With that said, the yield curve remains positive (1.22 and 1.76 for 10Y-2Y and 10Y-3M respectively), suggesting that investors expect future growth. In this regime, investors are nervous and the market is volatile, but the underlying economy is still in good shape as suggested by the bond market.

Regime 3: Panic or Market Correction

Regime 3 represents extreme volatility and investor fear. The VIX spikes to 42, the SMA_diff drops down to -14, and the SMA_Ratio falls below 0.9, representing technical breakdown and severe negative momentum. Despite this, RSI remains relatively stable at 52, and both yield curves (10Y-2Y at 1.85 and 10Y-3M at 2.575) stay positive, suggesting that fundamentals remain supportive. As such, this regime possibly represents sharp market corrections or external shocks rather than full recessions, with high risk of sector dispersion and short-term market whipsaws.

Regime 4: Recovery and Stabilization

Regime 4 represents improving technical and sentiment conditions. The RSI recovers to 57, the SMA_Ratio and SMA_diff rebound back to bullish levels at 1 and 6 respectively, and the VIX falls to 15.6, indicating reduced fear. The yield curves for 10Y-2Y and 10Y-3M remain modestly positive at 1.05 and 1.54 respectively, indicating policy support. This regime aligns with early recovery phases, where sentiment is rebuilding and the market offers more opportunities for risk-taking.

Subject to the analysis above, each regime will take on the following labels:

GMM Cluster	Regime Label
0	Moderate Expansion
1	Bullish Risk-On
2	Choppy Recovery
3	Panic / Market Correction
4	Recovery and Stabilization

Table 2: GMM-Detected Regimes and their Associated Labels

4.2 Sector ETF Returns per Regime

The next part of the analysis looks into the sector returns within each regime. For each regime, the sector-level daily returns are aggregated and used to estimate annualized Sharpe ratios, which indicates the *risk-adjusted* strength of each sector under each regime.

The Sharpe ratio is given by:

$$\text{Sharpe Ratio} = \frac{\mu_{\text{annual}}}{\sigma_{\text{annual}}} \quad (4)$$

where μ_{annual} is the annualized mean return and σ_{annual} represent the annualized volatility. Essentially, higher Sharpe ratios mean better returns relative to their risk exposure during that particular regime. Taking this value for each ETF for each regime, the heatmap in Figure 5 is obtained:

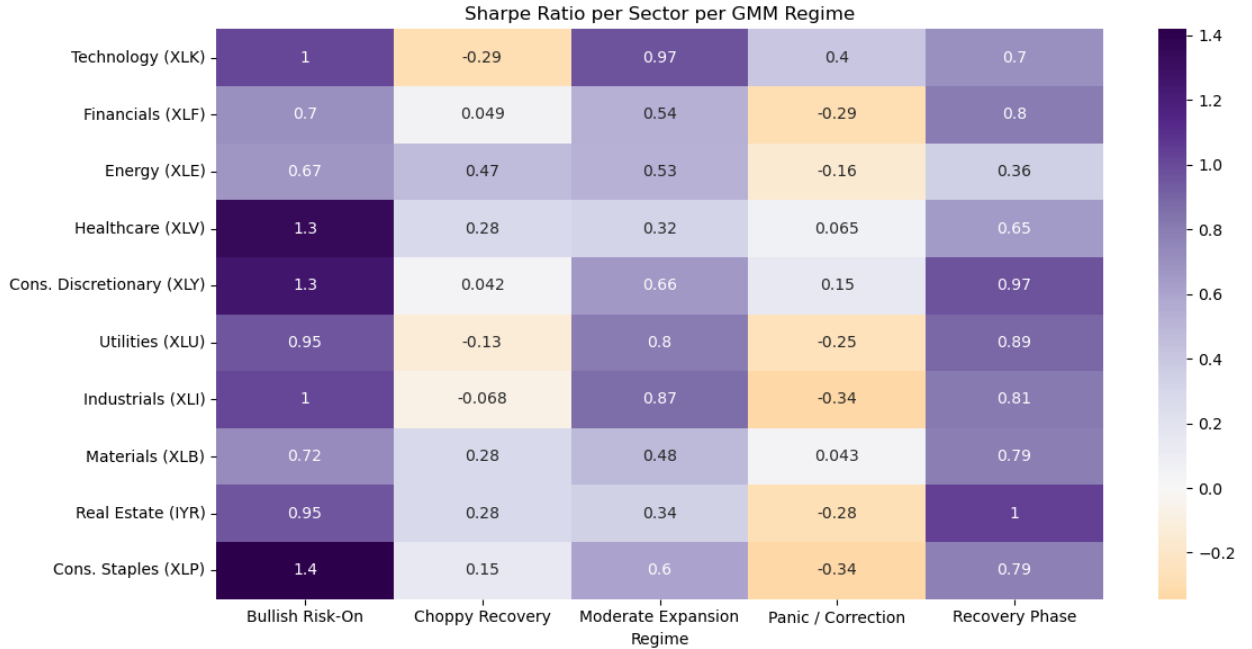


Figure 5: Sharpe Ratios by Sector Across GMM-Detected Regimes

In the visual above, Technology, Consumer Discretionary, and Consumer Staples consistently show strong Sharpe ratios during Bullish Risk-On regimes, while defensive ETFs like Utilities and Real Estate perform better during Recovery and Moderate Expansion regimes. During the Panic / Correction regime, most sectors exhibit low to negative Sharpe ratios. These sector-specific patterns will assist with the subsequent regime-based allocation strategy.

5 Sector Rotation Strategy

Now that the market regimes have been identified and labeled accordingly, the next step is to define a sector rotation strategy that can adjust dynamically to regime changes. Under each regime, the goal is to overweight sectors that historically outperformed under similar conditions and underweight or avoid those that underperformed.

Essentially, the sector rotation strategy is constructed by ranking sectors within each regime according to their Sharpe ratios and assigning portfolio weights based on this ranking. After assigning preliminary weights based on Sharpe ranking tiers, the weights are normalized within each regime to ensure that the total allocation sums to 100%.

To reduce overfitting and overreacting to small differences in Sharpe ratios, a bucketing approach is implemented instead of directly normalizing Sharpe ratios into weights. For each regime, all sector ETFs (including cash when applicable) are ranked by their Sharpe ratios, and allocations are assigned based on fixed tiers:

- The **top 3 sectors** receive **70% of the portfolio** (split as 24%, 23%, and 23%),
- The **next 3 sectors** receive **30%** (10% each),
- The **remaining sectors** are assigned zero weight.

Recall that a synthetic “cash” return feature was engineered from short-term Treasury yields to represent risk-free returns. In non-panic regimes, the allocation of cash is manually restricted to 0, ensuring complete exposure to the allocated sectors. However, during the Panic/Correction regime, cash is included alongside sectors based on its Sharpe ranking, providing a way to partially manage risk during this regime.

Next, to ensure that all weights sum to 100%, the weights are normalized within each regime, such that:

$$\text{Final Weight}_i = \frac{\text{Assigned Weight}_i}{\sum_j \text{Assigned Weight}_j} \quad (5)$$

Thus:

$$\sum_{i=1}^n \text{Final Weight}_i = 1 \quad (6)$$

Where n is the number of sectors (including cash during Panic/Correction regime). Using this approach, the normalized sector weights for each regime are summarized in Table 3, which will be used to dynamically rotate the sectors:

Sector	Bullish Risk-On	Choppy Recovery	Moderate Expansion	Panic / Correction	Recovery Phase
XLK_ret	0.10	0.00	0.24	0.23	0.00
XLF_ret	0.00	0.10	0.10	0.10	0.10
XLE_ret	0.00	0.00	0.24	0.00	0.00
XLV_ret	0.23	0.23	0.00	0.00	0.00
XLY_ret	0.23	0.00	0.10	0.23	0.23
XLU_ret	0.10	0.00	0.23	0.00	0.23
XLI_ret	0.10	0.00	0.23	0.00	0.20
XLB_ret	0.10	0.00	0.00	0.00	0.10
IYR_ret	0.00	0.23	0.00	0.00	0.24
XLP_ret	0.24	0.10	0.10	0.00	0.00
CASH_ret	0.00	0.00	0.00	0.24	0.00

Table 3: Regime-Specific Portfolio Allocation Weights

6 Evaluation and Final Results

To evaluate the effectiveness of the regime-based sector rotation strategy, a test is conducted using an initial capital of \$100,000. The portfolio dynamically reallocates the sector ETFs daily based on the detected market regime and the allocation weights derived in the previous section (Table 3).

For each day from 2000 to 2024, the model obtains the detected regime, applies the allocation weights for that regime, and updates the portfolio value multiplicatively based on the realized daily returns of those underlying sectors. For comparison purposes, a passive investment in the S&P 500 index is also tracked along with the GMM-driven portfolio. This setup allows for a fully controlled comparison of how the dynamic sector rotation strategy compares against a static buy-and-hold benchmark during this time period.

Mathematically, the portfolio return is computed as the weighted sum of sector returns:

$$r_t = \sum_{i=1}^n w_{i,t} \times r_{i,t} \quad (7)$$

Where $w_{i,t}$ is the portfolio weight assigned to sector i under the current regime, and $r_{i,t}$ is the realized daily return of sector i .

For each day t , the portfolio value is updated based on:

$$V_t = V_{t-1} \times (1 + r_t) \quad (8)$$

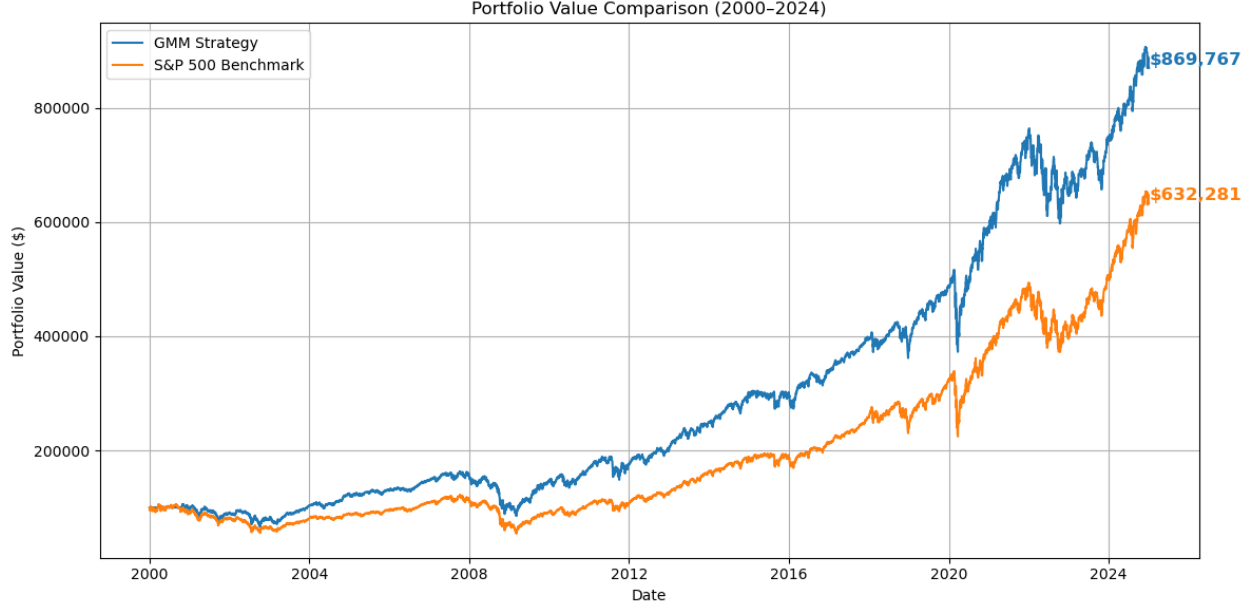


Figure 6: GMM Strategy vs. S&P 500 Benchmark (2000–2024)

Figure 6 shows the cumulative portfolio value of the GMM sector rotation strategy compared to the buy-and-hold S&P 500 benchmark, both starting with an investment of \$100,000. As per the visual, the GMM strategy outperforms the S&P 500, reaching \$869,767 compared to the S&P 500's \$632,281, which represents a 37.5% higher ending value.

Despite this, Figure 6 shows that the GMM strategy exhibits a high correlation with the overall S&P 500 index, tracking similar trends and patterns throughout the time series. But this is expected since the strategy is mostly allocated to equities and reacts to the same macroeconomic forces that drive the S&P 500, i.e. interest rate changes, economic cycles, and volatility. Additionally, the S&P 500 itself is actually composed of many of the same sector ETFs included in the rotation strategy, so the overall performance will naturally align.

Overall, it is evident that the GMM strategy delivers more growth over time, which reflects the model's ability to dynamically rotate allocations away from riskier sectors (or into cash) based on the different regimes. Despite experiencing downturns during known periods of market stress, the 37.5% higher end value speaks to how the GMM drive rotation strategy provides better compound growth and risk-adjusted performance compared to a static S&P 500 investment.

7 Conclusion

In conclusion, this study demonstrates that classifying market regimes using Gaussian Mixture Models provides a strong framework for sector-based adaptive portfolio management. By combining financial and macroeconomic data, the model effectively clustered bull, bear, recovery, and panic regimes from 2000 to 2024. When compared against a static S&P 500

benchmark, the GMM-driven sector rotation strategy produced superior risk-adjusted returns. Though the results are promising, ongoing refinement and testing are needed to validate the model’s true effectiveness in detecting regime shifts.

Future work will involve applying it to future data over the coming months and years. Monitoring performance across new market cycles will help assess the durability and adaptability of the regime classification and rotation strategy. Another area for improvement includes experimenting with other clustering algorithms or imposing fairness constraints on the GMM, such as the Fair Gaussian Mixture Model. These measures may reduce the model’s sensitivity to overlapping distributions. Lastly, early detection could be further enhanced by incorporating change detection methods like CUSUM or One-Class SVM. These methods could enable an investment strategy to respond more proactively to evolving market dynamics.